- *Personal assistant*—In most cases, an evaluation component will be used to process and guide the performance of agent responses. In systems primarily employing retrieval augmented generation (RAG) for document exploration, the evaluation indicates how well the assistant responds to information requests.
- *Customer service bot*—Evaluating service bots is critical to understanding how well the bot responds to customer requests. In many cases, a strong RAG knowledge element may be an element of the system that will require extensive and ongoing evaluation. Again, with most evaluation components, this element is external to the main working system and is often run as part of monitoring general performance over several metrics.
- *Autonomous agent*—In most cases, a manual review of agent output will be a primary guide to the success of an autonomous agent. However, in some cases, internal evaluation can help guide the agent when it's undertaking complex tasks or as a means of improving the final output. Multiple agent systems, such as CrewAI and AutoGen, are examples of autonomous agents that use internal feedback to improve the generated output.
- *Collaborative workflows*—In most direct cases, manual evaluation is ongoing within these types of workflows. A user will often immediately and in near real time correct the assistant/agent by evaluating the output. Additional agents could be added similarly to autonomous agents for more extensive collaborative workflows.
- *Game AI*—Evaluation will often be broken down into development evaluation—evaluating how the agent interacts with the game—and in-game evaluation, evaluating how well an agent succeeded at a task. Implementing the later evaluation form is similar to autonomous agents but aims to improve some strategies or execution. Such in-game evaluations would also likely benefit from memory and a means of feedback.
- *Research*—Evaluation at this level generally occurs as a manual effort after completing the research task. An agent could employ some form of evaluation similar to autonomous agents to improve the generated output, perhaps even contemplating internally how evaluation of the output could be extended or further researched. Because this is currently